

Information loss of fixed health indicators as a control problem: simplex geometry and reachability limits

Huy Hoang Le^a, Kim-Anh Nguyen^{b,*}

^a*R&D Department, PowerMore Ltd., Da Nang, 550000, Viet Nam*

^b*Faculty of Electrical Engineering, University of Science and Technology, The University of Danang, Da Nang, 550000, Viet Nam*

Abstract

A health indicator compresses several condition-monitoring channels into one scalar, and normally keeps the same weights for the life of the component. Because degradation runs through several mechanisms at differing rates, the channel combination carrying most information about remaining life rotates as the component ages, so a fixed indicator necessarily discards information. We show that the operating point sets how much it discards, which makes that loss a control variable. The loss reduces to a single angle, between the indicator and the direction along which information currently arrives, and writing each channel's share of that information as a probability vector places that angle on a statistical simplex. Zero loss is attainable exactly when the directions reachable at each instant share a common point, which for generic plants needs one fewer independently adjustable operating variable than there are mechanisms. Scaling every channel by the same factor leaves the loss unchanged, so no scalar summary of the Fisher information matrix can detect it. On a representative three-mechanism plant, steering the operating point cuts the discarded information by a factor of 68, and a dimensionless index, available before any indicator is chosen, says whether such a gain is plausible. Six public run-to-failure fleets test the premises separately: the informative direction does rotate on measured cells, and simulated turbofan records confirm against a negative control that the operating point moves it. The same fleets bound the theory: the highest-fidelity turbofan

*Corresponding author

Email addresses: `lhhoang@powermore.vn` (Huy Hoang Le), `nkanh@dut.udn.vn` (Kim-Anh Nguyen)

data reject the separable plant form under which the operator’s authority becomes computable offline, so that branch describes a model class rather than measured machinery.

Keywords: prognostics, health indicator, remaining useful life, information loss, optimal control, reachability

1. Introduction

Condition monitoring rarely feeds raw measurements to a life-prediction algorithm. A vibration channel, a temperature, an impedance and a capacity reading first go into a single scalar, the health indicator, and the life-prediction algorithm sees only that scalar [1, 2]. The compression is almost always fixed: the designer picks one set of weights at design time, and every unit carries those weights for its whole life, because firmware embeds them, certification freezes them, or re-tuning them in the field is impractical.

Fixing the compression has a cost. Real components degrade through several mechanisms at once, and those mechanisms do not proceed in step: an early-life process may dominate for the first third of the record and then saturate, while a late-life process accelerates. The combination of channels that carries the most information about remaining life therefore rotates as the component ages. A fixed indicator is aligned with that combination at best at one moment, and is misaligned for the rest of the record; the information it discards is not recoverable downstream, however good the prediction algorithm that follows [3].

The established response is to make the map better: choose a different criterion for it, let it switch between segments of life, or learn it from more run-to-failure data [4, 2]. All of these take the degradation as given and work on the compression. This paper acts on the degradation itself, through the operating point.

The rotation is not a property of the instrumentation. The degradation physics produces it, and that physics responds to stress: temperature, load and duty cycle change the relative rates of the competing mechanisms, not only the overall speed. Running a cell hotter does not only age it faster; it also shifts the balance between the mechanisms, and therefore moves the direction in which information about remaining life arrives.

The operating point is, in most applications, something the operator chooses. So the misalignment between a fixed indicator and the informa-

31 tive direction is not fixed by nature. It is a function of the input, and the
32 operator can steer it.

33 That reframing turns a data-analysis question into a control question, and
34 control questions come with sharper answers than design heuristics do. One
35 can ask what misalignments are reachable at a given operating envelope,
36 whether zero loss is attainable at all, how many independent inputs that
37 would take, and what the best achievable loss is when it is not. These are
38 the questions this paper answers.

39 They sit against five bodies of work. Four address parts of the problem
40 and the fifth supplies the geometry used in Section 3; to our knowledge none
41 of them addresses the position taken here.

42 The prognostics literature treats the compression as a design problem to
43 be solved from data: searching genetically over candidate features against
44 monotonicity and trendability criteria [2], learning it with a recurrent net-
45 work [5], or surveying the resulting pipelines [1, 6]. Two limits on that
46 programme are known, information-theoretic bounds on what any fixed com-
47 pression can retain [3] and the number of run-to-failure units needed to learn
48 one reliably [4]. The first is our own, and the boundary is worth drawing
49 explicitly: it takes one degradation trajectory as given and asks how much
50 information a fixed compression must lose along it. The question here is the
51 complementary one, whether that trajectory can itself be steered so that the
52 same fixed map loses less. Both prior limits hold the trajectory fixed; neither
53 asks whether it can be changed.

54 The predictors downstream are overwhelmingly stochastic — Wiener,
55 gamma and inverse-Gaussian degradation processes with first-passage fail-
56 ure [7, 8], reviewed for remaining-life estimation in [9, 10] and embedded in
57 system-level design methodology in [11]. That literature models how damage
58 evolves, not how the measurement should be compressed before the model is
59 applied, which is the question upstream of it.

60 A separate literature does treat the operating point as a decision variable,
61 but to extend remaining life rather than to preserve estimability [12, 13];
62 the two aims can conflict, and Section 6 exhibits an instance in which they
63 disagree. Choosing inputs to make a state easier to estimate is itself classical,
64 going back to dual control [14], and its modern forms shape a scalarisation
65 of the Fisher information matrix [15, 16, 17]. Our Proposition 4.6 shows the
66 quantity at issue here to be invisible to every such scalarisation.

67 The fifth body is information geometry. The reformulation of Section 3
68 rests on the Fisher–Rao geometry of the probability simplex and on the

Bhattacharyya coefficient [18, 19], and the flow it produces is the replicator equation of evolutionary dynamics [20, 21]. These are standard objects; what is new here is their identification with the informative direction of a degrading component, and the fact that the fitness driving the flow is actuated.

The contributions are as follows.

1. The information a fixed indicator discards is an exact function of one misalignment angle (Lemma 2.2). Placing the channels' shares of the Fisher information on a statistical simplex turns the degradation into a replicator flow whose rates the operating point modulates, and the optimal indicator into a barycentre of the degradation path (Section 3).
2. Zero loss is attainable exactly when the reachable direction sets share a point, which generically needs at least as many independent operating variables as mechanisms less one (Section 4). Supplying the missing input collapses the peak reachability floor from 4.52° to zero at every state, so the rule prices instrumentation rather than only bounding it. Taking logarithms makes every reachable set a translate of one fixed set, which turns the question into a containment and shows the counting rule to be sufficient but not necessary; the exceptions exist for any number of mechanisms and are too fragile to build on (Section 4.1). The count separates two objectives that are easily conflated: holding the misalignment at a constant level is a single scalar condition and generically needs one input, whereas freezing the informative direction outright is $d - 1$ conditions. That gap is the difference between useful steering and lossless steering, and an installation with one controllable variable can have the first without the second.
3. The loss is invariant to a common gain on all channels and is therefore invisible to every scalarisation of the Fisher information matrix, so observability-driven criteria cannot be used to choose an indicator.
4. The best achievable loss is the value of an optimal control problem whose objective is a ratio, and whose terminal time is itself controlled (Section 5), and a dimensionless regime index, computable before any indicator is designed, screens for the regime in which steering can have substantial value (Section 6.7).
5. The premises the construction rests on are separately testable on public run-to-failure data, and Section 7 tests them in three stages: whether the informative direction rotates at all, whether the operating point moves it, and whether the separable stress model survives. Two survive

and one does not, which makes that section a test of which hypotheses withstand contact with data rather than a demonstration that the method works. The rotation is present in measured fleets; the actuation premise is so far confirmed on simulation only; and higher-fidelity turbofan data reject the separability of Lemma 4.3, which places that lemma, and the offline route it opens, as statements about a model class rather than about measured machinery.

Section 2 states the model. Sections 3 to 5 develop the theory, Section 6 exercises the three of its claims that are stated sharply enough to be refuted, and Section 7 reports what public fleet data can and cannot establish about the premises the theory rests on.

2. Problem formulation

This section states the degradation model, says precisely what a prognostic algorithm is being asked to estimate and where the uncertainty in that quantity comes from, derives the direction along which information about it arrives, and defines the fixed indicator whose information loss the rest of the paper studies. Readers familiar with [3] will recognise the model; the derivation is given in full here because the geometry of Section 3 depends on several of its details.

2.1. Degradation driven by a controlled operating point

A component degrades through d distinct mechanisms acting at once. In a lithium-ion cell these might be loss of cyclable lithium, growth of the solid-electrolyte interphase and mechanical fatigue of the electrode; in a rolling bearing, spalling, wear and lubricant degradation. We write $x_i(t) \geq 0$ for the damage accumulated by mechanism i up to time t , and collect these into the state $x = (x_1, \dots, x_d)$.

One externally chosen operating point drives all d mechanisms. It collects the temperature, the mechanical or electrical load, and the duty cycle at which the operator runs the component. We write that operating point as

$$u(t) \in \mathcal{U} \subset \mathbb{R}^m, \quad (1)$$

with $\mathcal{U}$ compact, representing the operating envelope the application permits. Throughout this paper u is the control input: it is the decision variable, and the question we ask is what choosing it well can buy. The number of

138 independent inputs, m , turns out to decide what is achievable, and it is
 139 usually much smaller than the number of mechanisms d .
 140 The damage states evolve according to

$$\dot{x}_i = \varphi_i(u) h_i(x_i), \quad i = 1, \dots, d, \quad (2)$$

141 in which the two factors play different roles:

- 142 • $\varphi_i(u) > 0$ is the stress-acceleration factor of mechanism i : how much
 143 faster that mechanism runs at operating point u than at a reference
 144 point. It carries all the dependence on the input.
- 145 • $h_i(x_i) > 0$ is the intrinsic shape of mechanism i : how its rate depends
 146 on the damage it has already done. It carries all the dependence on
 147 the state.

148 This separation is the standard accelerated-life form and covers the stress
 149 models in routine use [22]. In the Arrhenius model the input is absolute
 150 temperature and $\varphi_i(u) = A_i \exp(-E_i/(ku))$, with $A_i > 0$ a pre-exponential
 151 constant, $E_i > 0$ the activation energy of mechanism i and k Boltzmann's
 152 constant. In the inverse-power-law model the input is a mechanical or electri-
 153 cal stress and $\varphi_i(u) = \kappa_i u^{n_i}$, with $\kappa_i > 0$ a scale factor and $n_i > 0$ the stress
 154 exponent. Choosing h_i recovers the exponential, power-law and stretched-
 155 exponential degradation classes [3]; the results below use only $\varphi_i > 0$ and
 156 $h_i > 0$.

157 The within-unit dynamics are deterministic, where most of the degrada-
 158 tion literature places the randomness inside a single unit's path [7, 8, 10].
 159 We put it between units instead (Section 2.2). The object of study is the
 160 direction along which information arrives, which is fixed by the relative rates
 161 of the competing mechanisms: a within-unit diffusion would jitter that di-
 162 rection without altering its systematic motion, whereas dispersion in the rate
 163 parameters across a population is what makes remaining life uncertain and
 164 therefore worth estimating.

165 We measure damage on a scale normalised so that mechanism i exhausts
 166 the component's tolerance when x_i reaches unity. A component is withdrawn
 167 at a condemnation limit at or below that point, so we write the end of life
 168 as the first passage of any one mechanism to a threshold $x_f \in (0, 1]$:

$$T_f = \inf\{t \geq 0 : \max_i x_i(t) \geq x_f\}. \quad (3)$$

169 No result below depends on the value of x_f , which enters only by fixing where
 170 a trajectory stops. The computed quantities of Section 6 do depend on it,
 171 and the value used there is stated with the rest of the test plant.

172 2.2. What is to be estimated, and why it is uncertain

173 The quantity a prognostic system must report is the remaining useful life

$$R(t) := T_f - t, \quad (4)$$

174 the time left before failure.

175 We must say where the uncertainty in R lies, because (2) is a deterministic
 176 differential equation: for one component whose initial damage and parame-
 177 ters were known exactly, T_f and hence R would be known exactly, and there
 178 would be nothing to estimate. The uncertainty is between components, not
 179 within one. Units built to the same specification differ in initial damage and
 180 in the coefficients of (2), which is the unit-to-unit variability that reliability
 181 engineering treats as the primary source of scatter in life data [22]. A prog-
 182 nostic algorithm observes one unit drawn from that population and must
 183 infer its remaining life.

184 We therefore treat R , at any given observation instant, as a random
 185 variable over the population of units, with prior variance $\text{Var}[R] > 0$. This
 186 variance is the uncertainty the algorithm exists to reduce, and it is the reason
 187 the estimation problem is not trivial. Everything below is written for a
 188 nominal trajectory of (2), with the fleet spread entering only through $\text{Var}[R]$.

189 2.3. Observations and the informative direction

190 Each damage state reaches the record through its own noisy channel,

$$y_i = x_i + \sigma_i \epsilon_i, \quad i = 1, \dots, d, \quad (5)$$

191 with $\sigma_i > 0$ the noise level of channel i and ϵ_i independent standard normal
 192 variables. The record available to the algorithm is the collection of these
 193 measurements over the observation window.

194 Equation (5) pairs each mechanism with a channel of its own. Condition
 195 monitoring in general returns $y = H(x, u) + \epsilon$, with sensors that mix mech-
 196 anisms, and the pairing is an assumption rather than a description. It is
 197 load-bearing: the positivity of $\mathbf{d}$ that Remark 3.1 shows the results to need
 198 follows from $\varphi_i > 0$ and $h_i > 0$ in these coordinates and need not survive

199 an arbitrary H . Where H is known, invertible and positivity-preserving the
 200 construction carries over to $H^{-1}y$; the general case does not reduce to it.

201 Two assumptions carry the sensitivity, and both are stated here rather
 202 than left to be inferred.

203 *The prior on R .* We take R to be Gaussian over the population with variance
 204 $\text{Var}[R]$. Linearising the record about the nominal trajectory, following [3],
 205 then makes the record and R jointly Gaussian, and the mutual informations
 206 below are exact. Linearisation alone would not do it: with a non-Gaussian
 207 prior of the same variance the Gaussian value is an upper bound on what
 208 a channel carries, and Lemma 2.2 becomes an inequality rather than an
 209 identity.

210 *How units differ.* We take the population spread to be a reparametrisation
 211 of a single degradation path: units differ in where they sit on the trajectory
 212 of (2), not in its shape. This is what makes R a coordinate along that path,
 213 and it is what the next display uses. Dispersion in the coefficients of (2) is a
 214 different matter, because it moves $\partial x/\partial \theta$ rather than sliding along $\partial x/\partial R$,
 215 and Remark 2.1 records what it costs.

216 The sensitivity of channel i to remaining life then follows from (4): be-
 217 cause R falls at unit rate as t advances,

$$\frac{\partial x_i}{\partial R} = -\frac{\partial x_i}{\partial t} = -\varphi_i(u) h_i(x_i). \quad (6)$$

218 **Remark 2.1** (what the time-shift assumption costs). *Equation (6) is exact*
 219 *when the population spread slides along one path, and only then. On the*
 220 *plant of Section 6.1, with units instead drawn by dispersing the degradation*
 221 *coefficients, the direction that the population regression of x on R actually*
 222 *returns stands about 45° from $-\varphi(u)h(x)$, across two families of dispersed*
 223 *coefficient and every spread tried; a control whose units are started on the*
 224 *reference trajectory, where (6) holds by construction, returns the same re-*
 225 *gression to within 0.2° . The misalignments this paper is about are a few*
 226 *degrees. Extending the construction to populations dispersed in θ , where the*
 227 *observation carries scatter that no value of R explains, is open.*

228 Dividing each channel by its own noise level puts them on a common
 229 scale, and taking magnitudes gives the whitened signal direction

$$\mathbf{d}(t) = \left[\sigma_i^{-1} \varphi_i(u) h_i(x_i) \right]_{i=1}^d, \quad \hat{\mathbf{d}} = \mathbf{d}/\|\mathbf{d}\|. \quad (7)$$

230 In words: $\mathbf{d}$ says how strongly, and in what combination of channels, the
 231 measurement responds to a change in remaining life. Its direction $\hat{\mathbf{d}}$ is the
 232 *informative direction*, the combination a health indicator has to be aligned
 233 with and the object named informally in Section 1. Its length $\|\mathbf{d}\|$ is the
 234 strength of that signal. The two objects carry the paper: $\|\mathbf{d}\|$ is how much
 235 information arrives and $\hat{\mathbf{d}}$ is where it arrives from, and the loss of a fixed
 236 indicator depends only on the second. Experiment design in the observability
 237 tradition scores the first, which is why Proposition 4.6 finds it blind to the
 238 loss.

239 We use two properties of (7) repeatedly below.

240 *Positivity.* Every entry of $\mathbf{d}$ is strictly positive, because $\varphi_i > 0$ and $h_i >$
 241 0 in (2). This is a consequence of the degradation model rather than an
 242 assumption imposed for convenience, and Remark 3.1 shows that the results
 243 below would fail without it.

244 *Actuation.* $\mathbf{d}$ is an explicit algebraic function of the state and the input.
 245 The direction along which information about R arrives is therefore not a
 246 fixed property of the instrumentation: it moves when the operating point
 247 moves. That is the observation this paper is built on, and Fig. 1 places it:
 248 the operating point reaches $\hat{\mathbf{d}}$ twice, once through the state and once directly.

249 Because R is scalar, a single signal-to-noise ratio governs the information
 250 the record carries about it, namely the squared sensitivity times the prior
 251 uncertainty,

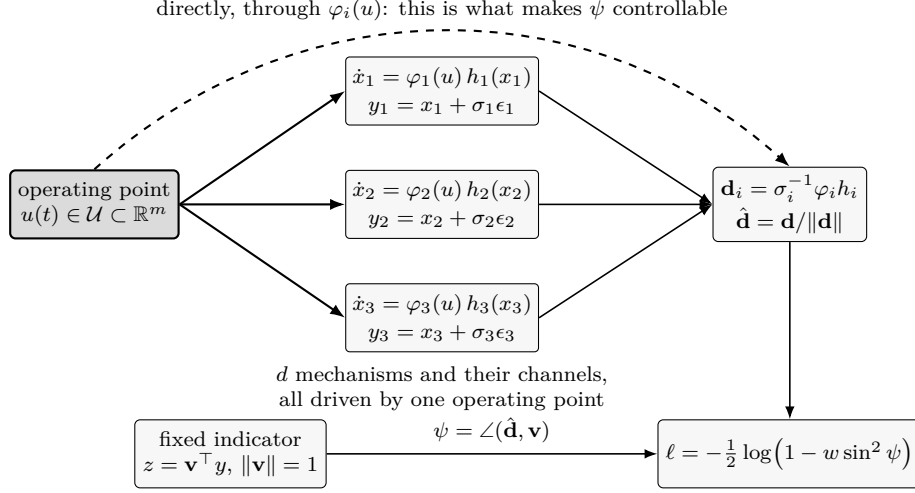
$$\gamma = \|\mathbf{d}\|^2 \text{Var}[R]. \quad (8)$$

252 Rates in (7) are taken with respect to clock time, whereas [3] differentiates
 253 with respect to age normalised by T_f . The two conventions differ by a factor
 254 common to all d channels, so they give the same $\hat{\mathbf{d}}$, and hence the same
 255 geometry in Sections 3–4, while rescaling γ . Section 5, where lifetime enters
 256 the objective, restores the normalised convention.

257 2.4. The fixed health indicator and its information loss

258 A health indicator compresses the d measured channels into one scalar
 259 before any life prediction is made. We take it to be a fixed linear map.
 260 Since (7) already measures each channel in units of its own noise, we write
 261 the indicator in those same coordinates: with $\Sigma := \text{diag}(\sigma_1^2, \dots, \sigma_d^2)$ and
 262 $\tilde{y} := \Sigma^{-1/2}y$ the whitened record,

$$z = \mathbf{v}^\top \tilde{y}, \quad \|\mathbf{v}\| = 1, \quad (9)$$



Alignment asks $\hat{\mathbf{d}}(x, u) = q$: $d - 1$ scalar equations on m unknowns, whence the counting rule of Theorem 4.1.

Figure 1: The construction. The operating point enters twice, through the plant and directly through $\varphi_i(u)$; the second path is what makes the misalignment ψ a control variable. One input drives all d mechanisms.

263 with the weight vector $\mathbf{v}$ chosen once, at design time, and held for the whole
 264 life of the component. Fixing it is not a modelling convenience: the map
 265 is typically embedded in firmware or frozen by certification, so it cannot be
 266 re-tuned as the component ages. An indicator specified on the raw measure-
 267 ments, $z = \mathbf{w}^\top \mathbf{y}$, is the same map under $\mathbf{v} \propto \Sigma^{1/2} \mathbf{w}$, and the normalisation
 268 $\|\mathbf{v}\| = 1$ only fixes the scale of z ; it restricts nothing. The distinction matters
 269 because the angle below is measured after whitening: unless the σ_i are equal,
 270 a unit vector in raw coordinates is not a unit vector in whitened ones, and
 271 the two readings of ψ disagree.

272 We compute the cost of fixing $\mathbf{v}$ as follows. Write $\psi(t)$ for the angle
 273 between the informative direction $\hat{\mathbf{d}}(t)$ and the indicator weights $\mathbf{v}$. Af-
 274 ter whitening, the signal in the record points along $\hat{\mathbf{d}}$ and has magnitude
 275 $\|\mathbf{d}\| \sqrt{\text{Var}[R]}$, while the noise is isotropic of unit scale. Projecting onto a unit
 276 vector $\mathbf{v}$ therefore keeps the fraction $\cos \psi$ of the signal and all of the noise,
 277 so the indicator's signal-to-noise ratio is

$$\gamma \cos^2 \psi \tag{10}$$

278 against the γ of (8) available to an oracle that is free to re-aim its projection at

279 $\hat{\mathbf{d}}(t)$ at every instant. Since a scalar Gaussian channel of signal-to-noise ratio
 280 s carries $\frac{1}{2} \log(1 + s)$ nats about R [23], the information the fixed indicator
 281 forfeits, at each instant, is

$$\ell = \frac{1}{2} \left[\log(1 + \gamma) - \log(1 + \gamma \cos^2 \psi) \right] \geq 0, \quad (11)$$

282 The design objective throughout is its average over the life of the component,

$$\bar{\ell} := \frac{1}{T_f} \int_0^{T_f} \ell(t) dt, \quad (12)$$

283 a loss per unit time. We optimise this rate rather than the information the
 284 whole observation record forfeits, because (12) is unaffected by how densely
 285 the record is sampled and is therefore a property of the indicator rather
 286 than of the measurement schedule. The two are different quantities, and
 287 Proposition 2.3 gives the second in closed form.

288 Two remarks on what (11) does and does not say. It is a loss relative to
 289 an oracle that no fixed map can realise, so it measures the price of the time-
 290 invariance constraint rather than of a bad design; a perfectly chosen $\mathbf{v}$ still
 291 pays it whenever $\hat{\mathbf{d}}$ moves. And it depends on the indicator only through the
 292 single angle ψ : two indicators that make the same angle with $\hat{\mathbf{d}}$ are equally
 293 good, whatever else distinguishes them.

294 The next lemma replaces (11) by a closed form that will be more conve-
 295 nient throughout, and records how it behaves at small misalignment.

296 **Lemma 2.2.** *Let $w := \gamma/(1 + \gamma) \in (0, 1)$. Then*

$$\ell = -\frac{1}{2} \log(1 - w \sin^2 \psi), \quad (13)$$

297 *exactly and at every signal-to-noise ratio, and*

$$\ell = \frac{w}{2} \psi^2 + \frac{w(3w-2)}{12} \psi^4 + O(\psi^6). \quad (14)$$

298 *Proof.* Substituting $\cos^2 \psi = 1 - \sin^2 \psi$ into (11) gives

$$\ell = -\frac{1}{2} \log \frac{1 + \gamma - \gamma \sin^2 \psi}{1 + \gamma} = -\frac{1}{2} \log(1 - w \sin^2 \psi).$$

299 For (14), use $-\frac{1}{2} \log(1-s) = \frac{s}{2} + \frac{s^2}{4} + O(s^3)$ with $s = w \sin^2 \psi$ and $\sin^2 \psi = \psi^2 -$
 300 $\psi^4/3 + O(\psi^6)$; collecting the ψ^4 terms gives $-w/6 + w^2/4 = w(3w-2)/12$. $\square$

301 The quadratic term of (14) is the bound obtained in [3, Thm. 1], which is
 302 therefore the leading term of an exact identity rather than an approximation
 303 of unknown quality. The quartic coefficient $w(3w - 2)/12$ changes sign at
 304 $w = 2/3$, that is at $\gamma = 2$: below that signal-to-noise ratio the quadratic
 305 model overstates the loss, above it the model understates it. This matters in
 306 practice because prognostic problems sit on both sides of $\gamma = 2$: an early-life
 307 measurement with a weak signal on one side, a well-instrumented component
 308 near failure on the other. Fig. 2(a) plots the exact loss against its quadratic
 309 approximation on both sides of that ratio.

310 **Proposition 2.3.** *Let the record consist of measurements (5) at instants*
 311 *$t_1, \dots, t_N$, with the noise independent across channels and across instants,*
 312 *and write $\mathbf{d}_k := \mathbf{d}(t_k)$ and ψ_k for the angle between $\hat{\mathbf{d}}_k$ and $\mathbf{v}$. Put*

$$G := \text{Var}[R] \sum_{k=1}^N \|\mathbf{d}_k\|^2, \quad W := \frac{G}{1+G},$$

$$S := \frac{\sum_k \|\mathbf{d}_k\|^2 \sin^2 \psi_k}{\sum_k \|\mathbf{d}_k\|^2}.$$

313 *Then the information about R that the fixed indicator forfeits over the whole*
 314 *record is*

$$\ell_{\text{rec}} = -\frac{1}{2} \log(1 - WS). \quad (15)$$

315 *Proof.* In whitened coordinates the linearised record is $\tilde{y}_k = \mathbf{d}_k R + \epsilon_k$ with
 316 $\epsilon_k \sim N(0, I_d)$ independent, so the Fisher information it carries about R is
 317 $\sum_k \mathbf{d}_k^\top \mathbf{d}_k = \sum_k \|\mathbf{d}_k\|^2$ and an oracle reading the whole record obtains $\frac{1}{2} \log(1 +$
 318 $G)$ nats. The fixed indicator returns the scalars $z_k = \mathbf{v}^\top \tilde{y}_k = (\mathbf{v}^\top \mathbf{d}_k)R + \mathbf{v}^\top \epsilon_k$,
 319 whose noise terms $\mathbf{v}^\top \epsilon_k$ are standard normal and independent because $\|\mathbf{v}\| =$
 320 1 , so the compressed record carries Fisher information $\sum_k (\mathbf{v}^\top \mathbf{d}_k)^2$. Since
 321 $\mathbf{v}^\top \mathbf{d}_k = \|\mathbf{d}_k\| \cos \psi_k$,

$$\text{Var}[R] \sum_k (\mathbf{v}^\top \mathbf{d}_k)^2 = \text{Var}[R] \sum_k \|\mathbf{d}_k\|^2 \cos^2 \psi_k = G(1 - S),$$

322 and subtracting the two mutual informations gives $\frac{1}{2} \log(1 + G) - \frac{1}{2} \log(1 +$
 323 $G(1 - S)) = -\frac{1}{2} \log \frac{1+G-GS}{1+G} = -\frac{1}{2} \log(1 - WS)$. $\square$

324 Equation (15) is (13) again, with the instantaneous signal-to-noise ratio
 325 γ replaced by its sum G over the record, and $\sin^2 \psi$ by the mean S weighted

by the signal energy $\|\mathbf{d}_k\|^2$. The reduction to one angle therefore survives the passage from an instant to a record. What does not survive is the averaging: $\bar{\ell}$ averages the loss, whereas (15) averages $\sin^2 \psi$ inside it, and the two differ. They also select different indicators. Minimising (15) over $\|\mathbf{v}\| = 1$ maximises $\mathbf{v}^\top (\sum_k \mathbf{d}_k \mathbf{d}_k^\top) \mathbf{v}$, so the best fixed indicator for the record is the leading eigenvector of the integrated Fisher matrix, while the minimiser of $\bar{\ell}$ is the barycentre of Proposition 3.3. Every result below is stated for $\bar{\ell}$.

Corollary 2.4 (what the loss costs an estimator). *In the model of Section 2, the minimum-variance estimate of R formed from the fixed indicator carries mean-squared error*

$$\frac{\text{Var}[R \mid z]}{\text{Var}[R \mid \tilde{y}]} = \frac{1 + \gamma}{1 + \gamma \cos^2 \psi} = e^{2\ell}$$

times that of the estimate formed from the whole whitened record, at each instant. The same identity holds over a record with ℓ replaced by ℓ_{rec} of Proposition 2.3.

Proof. Write $\tilde{y} = \mathbf{d}R + \epsilon$ with $\epsilon \sim N(0, I_d)$ and R Gaussian. The indicator returns $z = (\mathbf{v}^\top \mathbf{d})R + \mathbf{v}^\top \epsilon$, whose noise is standard normal because $\|\mathbf{v}\| = 1$, so its signal-to-noise ratio is $\gamma \cos^2 \psi$ and $\text{Var}[R \mid z] = \text{Var}[R]/(1 + \gamma \cos^2 \psi)$. The record's sufficient statistic is $\mathbf{d}^\top \tilde{y}$, with ratio γ , giving $\text{Var}[R \mid \tilde{y}] = \text{Var}[R]/(1 + \gamma)$. Their quotient is $e^{2\ell}$ by (11). The record statement is the same computation with G and S in place of γ and $\sin^2 \psi$. $\square$

The loss is therefore not a score standing in for estimation error: it is the logarithm of the factor by which that error inflates. What the corollary does not settle is whether the ordering survives an estimator other than the one the model prescribes, and Section 6.8 checks that separately.

3. The information simplex and its replicator flow

Lemma 2.2 reduced the loss to one angle. This section shows that the angle lives on a familiar object, the probability simplex, and that the degradation traces a flow along it that the operating point drives. The payoff is threefold: the loss becomes a recognised statistical distance, the optimal indicator becomes a barycentre, and the effect of the input becomes a single interpretable quantity. Two of the results here are load-bearing for what

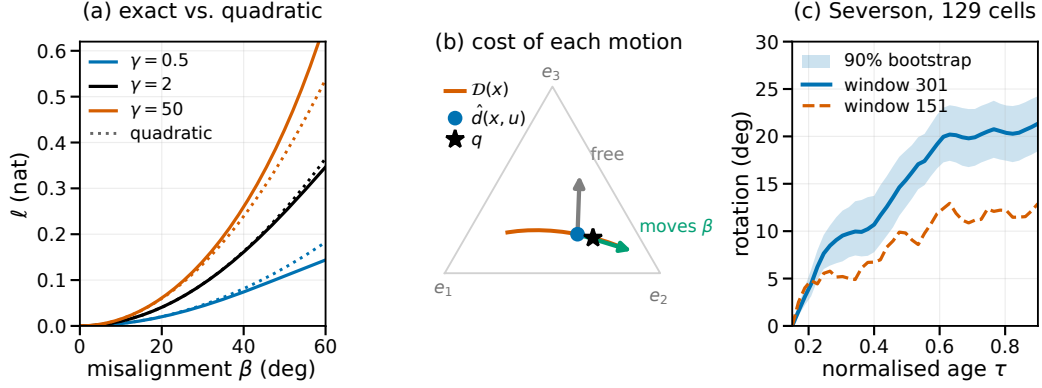


Figure 2: (a) Lemma 2.2: the exact loss (solid) against its quadratic model (dotted), whose quartic coefficient changes sign at $\gamma = 2$. (b) Theorem 3.6: of the replicator velocity, only the component along the geodesic towards q moves β . (c) The rotation, measured on the Severson fleet at two differentiation windows.

356 follows — the replicator form of Theorem 3.4 and the misalignment rate of
 357 Theorem 3.6, which Section 4 builds the reachability argument on. The Bhat-
 358 tacharyya identity and the barycentre are interpretation: they say what the
 359 loss and the optimal indicator already are in recognised terms, and nothing
 360 later depends on them.

361 3.1. Information shares

362 Because every entry of $\mathbf{d}$ is positive (Section 2.3), the squared components
 363 of the unit vector $\hat{\mathbf{d}}$ form a probability vector:

$$p_i := \hat{\mathbf{d}}_i^2, \quad p_i > 0, \quad \sum_i p_i = 1. \quad (16)$$

364 The interpretation is direct. The total Fisher information the record carries
 365 about R is $\|\mathbf{d}\|^2 \text{Var}[R]$ by (8), and channel i contributes $\mathbf{d}_i^2 \text{Var}[R]$ of it. So p_i
 366 is exactly the share of the total information about remaining life that arrives
 367 through channel i . A component in which one mechanism dominates the
 368 degradation has p near a vertex of the simplex; one in which three mecha-
 369 nisms contribute equally sits near its centre.

370 The same map applied to the indicator, $q_i := \mathbf{v}_i^2$, is legitimate because an
 371 optimal indicator may be taken with non-negative weights.

372 **Remark 3.1** (positivity is load-bearing). *If $\hat{\mathbf{d}} \geq 0$ then $\hat{\mathbf{d}}^\top |\mathbf{v}| \geq |\hat{\mathbf{d}}^\top \mathbf{v}|$*
 373 *at every instant, so replacing $\mathbf{v}$ by $|\mathbf{v}|$ never increases (11): non-negative*

374 *weights are without loss of generality. This argument uses the positivity of $\mathbf{d}$*
 375 *inherited from $\varphi_i, h_i > 0$, and it fails without it: in random trials with sign-*
 376 *indefinite $\hat{\mathbf{d}}$, taking absolute values was worse than the signed vector about a*
 377 *third of the time. The results therefore require the positive-orthant structure*
 378 *of the degradation model; it is not only a notational convenience.*

379 3.2. The loss as a Bhattacharyya distance

380 The identification below is used twice in what follows: it gives “the aver-
 381 age of a curve on the simplex” an unambiguous meaning (Section 3.3), and
 382 it lets the high-signal limit of the loss be read off as a known divergence.

383 **Proposition 3.2.** *$\cos \psi = \sum_i \sqrt{p_i q_i}$ is the Bhattacharyya coefficient of p and*
 384 *q . Consequently $\psi = \frac{1}{2} d_{\text{FR}}(p, q)$, where d_{FR} is the Fisher–Rao distance on*
 385 *the simplex, and, writing $D_{\text{B}}(p, q) = -\log \sum_i \sqrt{p_i q_i}$ for the Bhattacharyya*
 386 *divergence,*

$$\ell = -\frac{1}{2} \log \left[(1 - w) + w e^{-2D_{\text{B}}(p, q)} \right]. \quad (17)$$

387 *As $\gamma \rightarrow \infty$, $\ell \rightarrow D_{\text{B}}(p, q)$.*

388 *Proof.* By construction,

$$\cos \psi = \hat{\mathbf{d}}^\top \mathbf{v} = \sum_i \sqrt{p_i} \sqrt{q_i},$$

389 which is precisely the Bhattacharyya coefficient of p and q . The map $p \mapsto \sqrt{p}$
 390 is the classical isometry between the simplex under the Fisher–Rao metric
 391 and the positive orthant of the unit sphere, under which the geodesic distance
 392 is twice the spherical angle [18, 19]; hence $\psi = \frac{1}{2} d_{\text{FR}}$. Then $\cos^2 \psi = e^{-2D_{\text{B}}}$,
 393 and substituting into (13) gives (17). Letting $w \rightarrow 1$ gives the limit. $\square$

394 The square root is a bijection between the simplex and the non-negative
 395 part of the unit sphere, carrying p to $\hat{\mathbf{d}}$ and q to $\mathbf{v}$, and by the proposition
 396 it is an isometry. We use the two descriptions interchangeably from here
 397 on: a statement such as $q \in \mathcal{D}(x)$ in Section 4 says that $\mathbf{v}$ is a reachable
 398 direction, and a distance between points of the simplex is the angle between
 399 the corresponding unit vectors.

400 The resulting picture is the one the rest of the paper develops. The
 401 degrading component is a curve on the simplex, the health indicator is a
 402 single point on it, and the information lost depends only on the distance
 403 between them. Design of the indicator is the choice of one point; control of
 404 the operating point is the shaping of the curve.

405 3.3. The optimal indicator as a barycentre

406 If the indicator is one point and the plant is a curve, the best point should
 407 be some kind of average of the curve. It is, but of two different kinds in two
 408 different limits.

409 **Proposition 3.3.** *Let q^* minimise the time-averaged loss.*

- 410 (i) *As $\gamma \rightarrow \infty$, $\ell \rightarrow D_B$ pointwise, so q^* tends to the Bhattacharyya*
 411 *barycentre $\arg \min_q \int D_B(p_\tau, q) d\tau$.*
 412 (ii) *As the curve shortens, $\ell = \frac{w}{2}\psi^2 + O(\psi^4)$, so q^* tends to the w -weighted*
 413 *Fisher–Rao barycentre, and $\bar{\ell} = \frac{1}{8}\bar{w} \text{Var}^{\text{FR}} + O(\psi^4)$ where $\text{Var}^{\text{FR}} :=$*
 414 *$\min_q \mathbb{E}_\mu[d_{\text{FR}}^2(p_\tau, q)]$ is the μ -weighted Fréchet variance of the curve.*

415 *Proof.* (i) As $\gamma \rightarrow \infty$ we have $w \rightarrow 1$, and (13) gives

$$\ell \rightarrow -\frac{1}{2} \log \cos^2 \psi = -\log \sum_i \sqrt{p_i q_i} = D_B$$

416 by Proposition 3.2. The convergence is uniform, which is what carries the
 417 minimisers with it. Positivity keeps p_τ interior, and $\sqrt{q_i} \geq q_i$ with $\sum_i q_i = 1$
 418 gives $\cos \psi = \sum_i \sqrt{p_i q_i} \geq \min_i \sqrt{p_i}$ for every q in the simplex, so $\cos \psi$ is
 419 bounded away from zero uniformly in q and in τ and D_B stays finite. Both
 420 objectives are continuous on a compact set and uniformly close on it, so
 421 their minimisers converge. (ii) By (14), $\bar{\ell} = \frac{1}{2}\bar{w} \mathbb{E}_\mu[\psi^2] + O(\psi^4)$ with $d\mu =$
 422 $w d\tau/\bar{w}$; minimising a μ -weighted mean square of a geodesic distance is the
 423 definition of the weighted Fréchet mean, and $\psi = \frac{1}{2}d_{\text{FR}}$ converts $\frac{1}{2}\bar{w} \mathbb{E}_\mu[\psi^2]$
 424 into $\frac{1}{8}\bar{w} \mathbb{E}_\mu[d_{\text{FR}}^2]$. $\square$

425 The two limits are taken in different parameters, signal-to-noise ratio in
 426 (i) and arc length in (ii), and they do not coincide. On a curve of angular
 427 span 82.7° the two barycentres lie 0.72° apart, and q^* crosses from the Fisher–
 428 Rao side to the Bhattacharyya side between 0 and 10 dB, reaching 3×10^{-5}
 429 degrees of the latter by 50 dB. For design purposes the practical consequence
 430 is mild: either barycentre of a nominal degradation path is a good starting
 431 indicator.

432 3.4. The replicator flow and the role of the input

433 We now ask how fast the informative direction moves and what the op-
 434 erating point does to that motion.

435 **Theorem 3.4.** *Let $g_i = \log \mathbf{d}_i$ denote the logarithmic rate of channel i .*
 436 *Along (2),*

$$\dot{\hat{\mathbf{d}}}_i = \hat{\mathbf{d}}_i(\dot{g}_i - \mathbb{E}_p[\dot{g}]), \quad (18)$$

437 *so the information shares obey the replicator equation of evolutionary dynam-*
 438 *ics, in which each share grows in proportion to the amount by which its own*
 439 *rate exceeds the average, the average being taken over the shares themselves:*

$$\dot{p}_i = 2p_i(\dot{g}_i - \mathbb{E}_p[\dot{g}]), \quad (19)$$

440 *and*

$$\|\dot{\hat{\mathbf{d}}}\| = \text{Std}_p[\dot{g}], \quad \dot{g} = \left[\varphi_i(u) h'_i(x_i) \right]_i + \nabla_u \log \varphi \cdot \dot{u}. \quad (20)$$

441 *Proof.* From $\mathbf{d}_i = e^{g_i}$ we have $\dot{\mathbf{d}}_i = \mathbf{d}_i \dot{g}_i$, so $\dot{\hat{\mathbf{d}}} = \|\mathbf{d}\|^{-1}(\dot{\mathbf{d}} - \hat{\mathbf{d}}(\hat{\mathbf{d}}^\top \dot{\mathbf{d}}))$ has
 442 components $\hat{\mathbf{d}}_i \dot{g}_i - \hat{\mathbf{d}}_i \sum_j \hat{\mathbf{d}}_j^2 \dot{g}_j$, which is (18); squaring and summing gives
 443 $\text{Var}_p[\dot{g}]$. Since $p_i = \hat{\mathbf{d}}_i^2$, $\dot{p}_i = 2\hat{\mathbf{d}}_i \dot{\hat{\mathbf{d}}}_i = 2p_i(\dot{g}_i - \mathbb{E}_p[\dot{g}])$. Differentiating $g_i =$
 444 $\log \varphi_i(u) + \log h_i(x_i) - \log \sigma_i$ along (2) gives $\dot{g}_i = \nabla_u \log \varphi_i \cdot \dot{u} + (h'_i/h_i)\varphi_i h_i$,
 445 and $(h'_i/h_i)\varphi_i h_i = \varphi_i h'_i$. $\square$

446 Equation (20) has a compact reading. The speed at which the informative
 447 direction rotates equals the information-weighted spread of the mechanisms'
 448 logarithmic rates. If all mechanisms accelerate at the same proportional rate
 449 the direction does not move at all, however fast the component is degrading;
 450 it is disagreement among the mechanisms that rotates it. The identity is the
 451 Fisher–Rao/Shahshahani geometry of the simplex combined with Fisher's
 452 fundamental theorem of natural selection [18, 20, 21]. What is new here is
 453 its identification with the informative direction of a degrading component,
 454 and, decisively, that the quantity playing the role of fitness, the logarithmic
 455 rate $\dot{g}_i$ of channel i , is actuated: the second summand in (20) is the operator's
 456 lever.

457 **Remark 3.5** (contrast with Oja's flow). *Oja's flow for online principal-*
 458 *component extraction has the same algebraic form; for a diagonal covariance*
 459 *it reduces to a replicator equation with fitness twice the eigenvalues [24]. The*
 460 *roles are opposite. There the flow is an algorithm, run to fixation in order*
 461 *to find a direction, with constant fitness. Here it is the plant, its fitness is*
 462 *time-varying and actuated, and the aim is to prevent fixation from carrying*
 463 *the direction away from a fixed indicator.*

Only one scalar matters for the loss. Fix the indicator q and write $\beta := \angle(\hat{\mathbf{d}}, \mathbf{v}) = \frac{1}{2}d_{\text{FR}}(p, q)$ for the misalignment. This is the angle ψ of Lemma 2.2 under a new name, and the change of name marks a change of role: ψ was a fixed angle between two directions, whereas $\beta = \beta(x_t, u_t)$ is followed along the trajectory and is the quantity the input acts on. Every loss expression below may therefore be read as a function of β .

Theorem 3.6. *Let the operating-point trajectory $u(\cdot)$ be absolutely continuous, and let $\hat{e}$ be the unit tangent at $\hat{\mathbf{d}}$ along the geodesic towards $\mathbf{v}$. Then $\dot{\beta} = -\langle \dot{\hat{\mathbf{d}}}, \hat{e} \rangle$, which is affine in the input rate: $\dot{\beta} = \alpha_0(x, u) + \alpha(x, u)^\top \dot{u}$. In particular $|\dot{\beta}| \leq \|\dot{\hat{\mathbf{d}}}\|$, with equality only for motion along that geodesic.*

Proof. $\cos \beta = \hat{\mathbf{d}}^\top \mathbf{v}$ gives $-\sin \beta \dot{\beta} = \dot{\hat{\mathbf{d}}}^\top \mathbf{v}$. Also $\hat{e} = (\mathbf{v} - \cos \beta \hat{\mathbf{d}}) / \sin \beta$ and $\dot{\hat{\mathbf{d}}}^\top \hat{\mathbf{d}} = 0$, so $\langle \dot{\hat{\mathbf{d}}}, \hat{e} \rangle = \dot{\hat{\mathbf{d}}}^\top \mathbf{v} / \sin \beta = -\dot{\beta}$. Affinity follows because $\dot{g}$ is affine in $\dot{u}$ by (20) and $\dot{\hat{\mathbf{d}}}$ is linear in $\dot{g}$. $\square$

The hypothesis is needed only to write $\dot{u}$; nothing in Section 4 depends on it, since the reachable sets (21) are defined pointwise in u and the limits below hold for every measurable admissible input. Theorem 3.6 is a local kinematic description of smooth steering, not a restriction on the controls the rest of the paper admits.

The component of the replicator velocity orthogonal to $\hat{e}$ is therefore free: the informative direction may rotate in a $(d-2)$ -dimensional subspace at no cost in information. Fig. 2(b) shows the decomposition at one state. This is the observation that separates the two objectives of the next section.

4. Reachability limits on the misalignment

Controlling the rate $\dot{\beta}$ is not the same as reaching a value of β . The distinction is the subject of this section, and it is what decides whether a fixed indicator can be made lossless at all.

For a state x , the set of informative directions the operator can produce by choosing the operating point is

$$\mathcal{D}(x) := \{\hat{\mathbf{d}}(x, u) : u \in \mathcal{U}\} \subset S^{d-1}, \quad (21)$$

the reachable direction set, and the closest the indicator can be brought at that state is the reachable misalignment floor

$$\beta_{\min}(x, q) := \text{dist}(q, \mathcal{D}(x)). \quad (22)$$

494 For a single input, $\mathcal{D}(x)$ is an arc traced out as u sweeps $\mathcal{U}$; its length measures
 495 the actuation authority available at x , and it shrinks to a point when the
 496 mechanisms respond identically to the input.

497 **Theorem 4.1.** *Fix an admissible input and let $x(\cdot)$ be the trajectory it gener-*
 498 *ates. Zero loss along that trajectory is attainable with the fixed indicator q if*
 499 *and only if $q \in \cap_t \mathcal{D}(x_t)$. Solving $\hat{\mathbf{d}}(x, u) = q$ for u imposes $d - 1$ independent*
 500 *scalar equations on m unknowns, so for generic data a common point exists*
 501 *only if $m \geq d - 1$. If $m < d - 1$, then for generic data $\beta_{\min} > 0$ along*
 502 *the trajectory; Remark 4.5 exhibits plants for which it fails. In every case*
 503 *$\bar{\ell} \geq \int \ell(\beta_{\min}(x_t, q)) dt / T_f$ for that trajectory.*

504 *Proof.* By Lemma 2.2, $\ell(\beta) = 0$ iff $\sin \beta = 0$, so $\bar{\ell} = 0$ iff $\beta_t = 0$ for almost
 505 every t , that is iff $\hat{\mathbf{d}}(x_t, u_t) = q$, that is iff $q \in \mathcal{D}(x_t)$ for every t . Solving
 506 $\hat{\mathbf{d}}(x, u) = q$ imposes $d - 1$ independent scalar equations, both sides being
 507 unit vectors, on m unknowns. If $m < d - 1$ then $\mathcal{D}(x)$ is the image of an
 508 m -dimensional set, hence of dimension at most m , and is a null subset of the
 509 $(d - 1)$ -dimensional sphere that a generic q misses; if $m \geq d - 1$ the implicit
 510 function theorem supplies a solution wherever the Jacobian has full rank.
 511 The inequality $\beta_t \geq \beta_{\min}(x_t, q)$ holds for every admissible input by definition
 512 of $\beta_{\min}$ as a minimum over $\mathcal{U}$, and is strict exactly where $q \notin \mathcal{D}(x_t)$, which
 513 for $m < d - 1$ is generic. Since ℓ is increasing on $[0, \pi/2]$, the bound follows
 514 on dividing by T_f . $\square$

515 This is a state-conditioned result. The bound is genuine, in the sense
 516 that no input violates a reachability constraint at the state it occupies, but
 517 it is conditional on the trajectory, which the input also steers. Optimising
 518 over the trajectory as well is deferred to Section 5, and the two levels are the
 519 structure of the argument: a state-conditioned limit here, a trajectory-level
 520 limit there.

521 The practical reading of Theorem 4.1 is a counting rule: a component
 522 degrading through d mechanisms generically needs $d - 1$ independently ad-
 523 justable operating variables before a fixed indicator can be kept perfectly
 524 aligned. Section 4.1 says what the qualifier covers. Three mechanisms and a
 525 single temperature setpoint will not do it. The three results of this section
 526 answer different questions, and it is worth separating them. Theorem 4.1
 527 characterises zero loss exactly, as membership of every reachable set, and
 528 then counts dimensions to say when a generic plant can meet it; the count is
 529 necessary for generic data and is not by itself a construction. Proposition 4.4

530 gives an exact criterion, necessary and sufficient, on the separable subclass
 531 alone. Remark 4.5 exhibits plants outside that subclass which attain zero
 532 loss with fewer inputs than the count asks, so the count is not necessary in
 533 general. But a weaker and much cheaper objective is available.

534 **Corollary 4.2** (what actuation is actually required). *Two objectives that are*
 535 *easily conflated carry different requirements.*

- 536 (i) *Driving $\bar{\ell}$ to zero is equivalent to freezing $\hat{\mathbf{d}}$ at q , and needs $d - 1$*
 537 *conditions on $\dot{u}$, hence generically $m \geq d - 1$.*
- 538 (ii) *Holding β at a constant level, not necessarily zero, is the single scalar*
 539 *equation $\alpha_0 + \alpha^\top \dot{u} = 0$, solvable whenever $\alpha \neq 0$ and the resulting $u(\cdot)$*
 540 *stays in $\mathcal{U}$; generically $m \geq 1$ suffices.*

541 *Proof.* (i) By Theorem 4.1, $\bar{\ell} = 0$ iff $\hat{\mathbf{d}}(x_t, u_t) = q$ for every t , so $\hat{\mathbf{d}}$ is constant;
 542 by Theorem 3.4 that forces $\dot{g}_i = \mathbb{E}_p[\dot{g}]$ for every i , which is $d - 1$ independent
 543 conditions on $\dot{u} \in \mathbb{R}^m$. (ii) is one equation in $\dot{u}$ by Theorem 3.6. The gap
 544 between them is the content of that theorem: $\dot{\beta}$ sees only the component of
 545 $\hat{\mathbf{d}}$ along $\hat{e}$, so the remaining $d - 2$ directions of rotation are unconstrained by
 546 (ii) and constrained by (i). $\square$

547 Minimising $\bar{\ell}$ is neither of these: it trades the level of β against the
 548 trajectory that carries it, and is the problem of Section 5.

549 4.1. The reachable sets are translates of one set

550 Theorem 4.1 counts equations against unknowns and hedges with
 551 “generic”. The hedge is not idle, and the following observation says exactly
 552 what it covers.

553 **Lemma 4.3** (separability). *Write $\xi(x, u)$ for the class of $\log \mathbf{d}(x, u)$ modulo*
 554 *the all-ones vector, which is what $\hat{\mathbf{d}}$ retains. Then*

$$\begin{aligned} \xi(x, u) &= a(x) + b(u), \\ a_i(x) &= \log \frac{h_i(x_i)}{\sigma_i}, \quad b_i(u) = \log \varphi_i(u), \end{aligned} \tag{23}$$

555 *both taken modulo the all-ones vector. Consequently every reachable direction*
 556 *set is the same set translated: $\mathcal{D}(x) = a(x) + \mathcal{B}$ with $\mathcal{B} := b(\mathcal{U})$ fixed.*

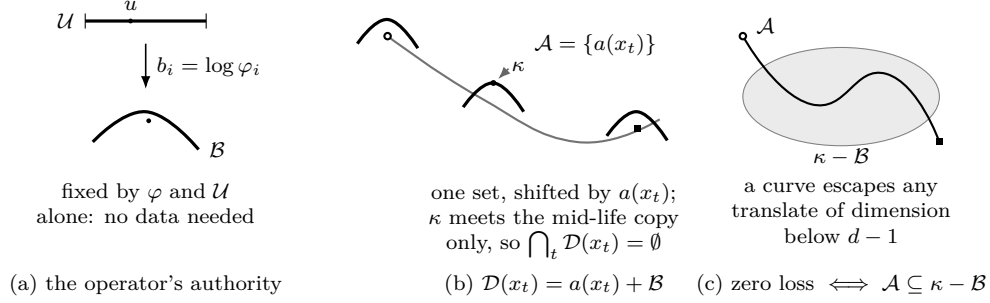


Figure 3: How the counting rule becomes a containment. (a) The operator's authority $\mathcal{B} = b(\mathcal{U})$, fixed by the stress model and the envelope alone. (b) Every reachable set is that one set translated, $\mathcal{D}(x_t) = a(x_t) + \mathcal{B}$ (circle: start of life, square: failure). (c) Zero loss asks the whole curve to lie in one translate (Proposition 4.4).

557 *Proof.* Immediate from $\mathbf{d}_i = \sigma_i^{-1} \varphi_i(u) h_i(x_i)$ in (7): the logarithm turns the
 558 product into a sum, and the normalisation of $\hat{\mathbf{d}}$ removes any component along
 559 the all-ones vector. $\square$

560 The decomposition separates the two parties. The set $\mathcal{B}$ is fixed by the
 561 stress model φ and the envelope $\mathcal{U}$ alone (Fig. 3(a)). An operator's authority
 562 can therefore be computed offline from the accelerated-life model, with no
 563 degradation record at all. The curve $\mathcal{A} := \{a(x_t)\}$ is fixed by the damage
 564 shapes h and the noise levels σ , and every reachable set is that one set carried
 565 along it (Fig. 3(b)). What the geometry of Section 3 measures is how far the
 566 plant's demand $\mathcal{A}$ sticks out of the operator's authority $\mathcal{B}$.

567 That offline route is the most useful consequence drawn in this paper
 568 and also the most restrictive. It holds on the separable subclass and not
 569 beyond it, and the best high-fidelity turbofan data we have reject separability
 570 (Section 7). This subsection and Proposition 4.4 should therefore be read as
 571 describing a structured subclass that is convenient when it applies, not as
 572 a general account of real machinery. Fig. 6 marks the one branch of the
 573 development that depends on it.

574 **Proposition 4.4** (exact criterion). *Zero loss is attainable along a trajectory*
 575 *if and only if $\mathcal{A}$ is contained in a translate of $-\mathcal{B}$. In particular $\mathcal{A}$ must lie*
 576 *in an affine subspace of dimension at most $\dim \mathcal{B} \leq m$, which for a curve*
 577 *in the $(d - 1)$ -dimensional quotient fails generically unless $m \geq d - 1$; this*
 578 *recovers Theorem 4.1. When it does not fail, $m < d - 1$ suffices.*

579 *Proof.* By Theorem 4.1, zero loss holds iff $q \in \mathcal{D}(x_t)$ for every t . Writing κ
580 for the quotient class of $\log \mathbf{v}$ and using Lemma 4.3, that reads $\kappa - a(x_t) \in \mathcal{B}$
581 for every t , that is $\mathcal{A} \subseteq \kappa - \mathcal{B}$. The dimension statement follows because a
582 translate of $\mathcal{B}$ spans an affine subspace of dimension $\dim \mathcal{B}$. $\square$

583 The containment is drawn in Fig. 3(c), and the mechanism the count
584 obscures is now visible. Alignment at an instant fixes u_t outright, since
585 $b(u_t) = \kappa - a(x_t)$ determines it; the input is spent, and any residual constraint
586 on the state must then hold along the flow with no freedom left to enforce
587 it. One input can carry the whole burden only when no residual constraint
588 is left, which happens when $\mathcal{A}$ already lies parallel to $\mathcal{B}$.

589 **Remark 4.5** (a family with $m = 1$ and any d). *Take exponential shapes*
590 $h_i(x) = e^{\nu_i(x_i - x_i^0)}$, *which (2) admits, and Arrhenius factors. Under the ramp*
591 $u(t) = u_0 + rt$ *the damage rates are constant, $x_i(t) = x_i^0 + \varphi_i^0 t$ with $\varphi_i^0 =$*
592 $\varphi_i(u_0)$, *so $a(x_t)$ moves at constant velocity $\nu_i \varphi_i^0$. Choosing $\nu_i = rE_i/\varphi_i^0$ makes*
593 *that velocity parallel to E , which is the direction of $\mathcal{B}$, and $b(u_t)$ retreats*
594 *along $\mathcal{B}$ at the same rate. The two cancel and $\hat{\mathbf{d}}$ is frozen. Integrating the*
595 *construction for $d = 2$ to 8 with one input holds the misalignment below*
596 1.5×10^{-6} *degrees and the mean loss below 10^{-15} . These families carry no*
597 *prior variance of their own, so their losses are evaluated at a fixed $w = 0.9$*
598 *rather than at a γ the plant supplies; the bound has fifteen orders of magnitude*
599 *of headroom and does not turn on the choice. That is against 1.2×10^{-5} to*
600 5.4×10^{-4} *for the best constant input on the same plant.*

601 This bears on the sharpness of Theorem 4.1. The family of Remark 4.5 is
602 a knife edge. Perturbing the shape coefficients ν by 1% raises the mean loss
603 to 3.2×10^{-4} under the myopic policy, against 1.8×10^{-4} for the best constant
604 input on the same perturbed plant. At one percent of shape error, steering
605 is already worse than not steering. Closing the loop halves the damage but
606 does not repair it, so the fragility belongs to the construction and not to the
607 open-loop schedule. Mistuning the activation energies instead is tolerated to
608 about two percent. A designer should read Proposition 4.4 as an explanation
609 of when the counting rule binds, and Theorem 4.1 as the rule to design
610 against.

611 4.2. Invisibility to Fisher-information criteria

612 A natural objection is that all of this should follow from designing the in-
613 put to maximise the Fisher information about R , as in the observability-aware
614 literature discussed in Section 1. It does not, and the reason is structural.

615 **Proposition 4.6** (Fisher blindness). *The loss of a fixed indicator is not a*
616 *function of the Fisher information about R : configurations with equal Fisher*
617 *information can have different losses, and configurations with equal losses*
618 *can have different Fisher information. No criterion depending on the Fisher*
619 *information alone can therefore rank fixed indicators by the loss they incur.*

620 *Proof.* For a scalar latent the Fisher information is the number $\|\mathbf{d}\|^2 \text{Var}[R]$,
621 so trace, determinant and least eigenvalue coincide and every scalarisation is
622 a strictly monotone function of it. Under the rescaling $\mathbf{d} \mapsto c\mathbf{d}$ with $c > 0$,
623 that number changes while $\hat{\mathbf{d}}$, and hence β and ℓ , do not. A common gain
624 therefore equalises the scores of two configurations with different losses, and
625 separates the scores of two configurations with the same loss. $\square$

626 Stated in engineering terms, a Fisher criterion measures the strength of
627 the signal, whereas the loss of a fixed indicator depends on its direction.
628 Amplifying every channel by the same factor increases the first and leaves
629 the second unchanged. Section 6 exhibits this concretely: with the Fisher
630 score held equal to seven digits, the loss still spans a factor of 9.4.

631 5. The best achievable loss as a fractional control problem

632 Theorem 4.1 bounds the loss along a given trajectory. But the input
633 steers the trajectory as well as the misalignment on it, so a fundamental
634 limit requires optimising over both at once. This changes the structure of
635 the problem.

636 The objective is a time average over a lifetime that the input itself deter-
637 mines:

$$\bar{\ell}^*(q) = \min_{u(\cdot) \in \mathcal{U}} \frac{\int_0^{T_f} \ell(\beta(x_t, u_t)) dt}{T_f}. \quad (24)$$

638 Both the integrand and the upper limit T_f depend on $u(\cdot)$. This makes
639 (24) a fractional optimal control problem, meaning that the objective is a
640 ratio rather than that the dynamics carry a fractional-order derivative. The
641 fraction is not a technicality: it is the source of the trade-off, as follows.

642 Suppose T_f were fixed. Nothing in (2) penalises the input directly, so
643 the answer would be trivial: find whichever admissible u minimises β and
644 stay there. Because T_f is itself controlled, the denominator of (24) rules that
645 solution out. The operating point that aligns the indicator is in general not
646 the operating point that prolongs life, so aligning now shortens the interval

over which the alignment is averaged. The optimum must trade the two, and Section 6.5 measures how steep that trade is on a concrete plant. A fixed-horizon formulation represents none of it, which is why one was tried and discarded (Section 6.5).

5.1. A converse bound uniform over trajectories

Before computing anything we record a certificate that removes the conditioning. Theorem 4.1 bounds the loss along a trajectory the input has already chosen; the following proposition bounds it over all of them at once, and it is what makes the number computed below a limit rather than an observation.

Proposition 5.1 (verification certificate). *Fix an indicator q and a constant $\lambda \geq 0$. Suppose there is a continuously differentiable V on a neighbourhood of the admissible state region such that*

- (i) $V(x) \leq 0$ for every x on the failure surface $\{\max_i x_i = 1\}$, and
- (ii) writing $D_i(x, u) = \varphi_i(u)h_i(x_i)$ for the drift of (2), for every admissible state x and every $u \in \mathcal{U}$,

$$\ell(\beta(x, u)) - \lambda + \nabla V(x)^\top D(x, u) \geq 0. \quad (25)$$

Then every admissible input satisfies $\bar{\ell} \geq \lambda + V(x_0)/T_f$. In particular, if $V(x_0) \geq 0$ then $\bar{\ell}^*(q) \geq \lambda$, uniformly over all admissible inputs and hence over all trajectories.

Proof. Let $u(\cdot)$ be any admissible input and x_t its trajectory. Since V is C^1 and $t \mapsto x_t$ is absolutely continuous, $t \mapsto V(x_t)$ is absolutely continuous and the chain rule gives $\frac{d}{dt}V(x_t) = \nabla V(x_t)^\top \dot{x}_t = \nabla V(x_t)^\top D(x_t, u_t)$ for almost every t . Applying (25) at (x_t, u_t) gives $\ell(\beta(x_t, u_t)) - \lambda + \frac{d}{dt}V(x_t) \geq 0$ a.e., and integrating over $[0, T_f]$,

$$\int_0^{T_f} (\ell - \lambda) dt \geq V(x_0) - V(x_{T_f}) \geq V(x_0),$$

the last step by (i), since x_{T_f} lies on the failure surface. Dividing by $T_f > 0$ gives the claim. $\square$

Condition (25) constrains the plant data alone: it quantifies over every state and every admissible input, and mentions no trajectory. Any V one

675 can exhibit therefore certifies a limit, and the search for a converse becomes
676 the search for a verification function.

677 The C^1 hypothesis cannot be weakened to Lipschitz continuity. A Lip-
678 schitz V is differentiable almost everywhere in the state by Rademacher’s
679 theorem, but a single trajectory is a null set and could in principle run inside
680 the exceptional set, so the chain rule step would be unavailable; recovering
681 the argument at that generality needs the Clarke generalised gradient in
682 place of ∇V . Smoothness is also, as it turns out, exactly what the natural
683 candidates lack.

684 We tried two. The value function of the Dinkelbach problem below sat-
685 isfies (25) with equality in the viscosity sense [25], but value functions of
686 this class are only Lipschitz. Computed on a 41^3 grid and tested on a 61^3
687 one, (25) fails at about 45% of test points by up to 2.9×10^{-3} . Halving λ
688 moves that violation by 0.6%, so the fault lies in the candidate and not in
689 the bound. At the opposite extreme an affine $V(x) = c_0 - a_0^\top x$ has an exactly
690 constant gradient and makes every requirement linear, so the best certifiable
691 λ is a linear program; over 167,936 constraints it returns $V \equiv 0$ and only
692 the pointwise bound $\min_{x,u} \ell$, which is vacuous here because the reachable
693 floor falls to within sampling resolution of zero at mid-life. Deleting the two
694 boundary requirements lets that program run to any cap imposed on it, which
695 identifies them as what binds: trajectories of (2) increase in every coordinate,
696 so an affine V is monotone along all of them at once and $V(x_0) \geq 0 \geq V|_{\text{failure}}$
697 forces it to decrease, while the near-aligned states, where the running cost
698 vanishes, demand a local increase.

699 The two failures are complementary, and together they say what a cer-
700 tificate needs. The value function varies its gradient exactly as the two de-
701 mands require but is not differentiable where that matters; affine functions
702 are perfectly smooth but cannot vary their gradient at all. A certificate must
703 therefore be both smooth and non-constant in its gradient, which points at
704 a smooth under-approximation of the value function, obtained by sum-of-
705 squares programming when the plant data are polynomial, or by interval
706 arithmetic with an explicit Lipschitz constant. We do not claim a certified
707 converse here, and the limit below remains a computation; what Proposi-
708 tion 5.1 buys is the replacement of an open problem without shape by a
709 concrete object with a checkable property.

710 Problems of the form (24) are handled by Dinkelbach’s transform [26]:

711 setting

$$F(\lambda) = \min_{u(\cdot) \in \mathcal{U}} \int_0^{T_f} (\ell - \lambda) dt, \quad (26)$$

712 the map F is decreasing in λ and vanishes exactly at the optimal ratio, so λ^* is
 713 found by bisection on $F(\lambda) = 0$. The inner problem in (26) has free terminal
 714 time and an absorbing set, hence a stationary Hamilton–Jacobi–Bellman
 715 equation on the state alone, with no time argument. Because $\dot{x} > 0$ the
 716 state advances monotonically, and a semi-Lagrangian scheme [27] converges
 717 by value iteration. One implementation detail matters: the scheme must use
 718 a fixed spatial step rather than a fixed time step, because degradation speeds
 719 vary by orders of magnitude across $\mathcal{U}$ and a fixed time step is unusable at
 720 one end of the envelope or the other.

721 6. Numerical study

722 Sections 3–5 are analytical, and three of their claims are stated sharply
 723 enough to be tested rather than illustrated: that the loss is invisible to a
 724 Fisher criterion, that the counting rule $m \geq d - 1$ binds in both direc-
 725 tions on a generic plant, and that the best achievable loss is a finite number
 726 one can compute. This section instantiates the model on a synthetic three-
 727 mechanism plant on which every quantity is available in closed form, checks
 728 those three claims against it, measures what steering is worth, and reads off
 729 the dimensionless index that says in advance whether it will repay the effort.
 730 Section 7 then reports what measured fleets can and cannot establish about
 731 the premises.

732 6.1. The test plant

733 We take $d = 3$ mechanisms with Arrhenius factors $\varphi_i(u) = e^{\theta_i - E_i u}$, in
 734 which the input u is the reciprocal temperature $1/(kT)$, so that this is the
 735 Arrhenius form of Section 2 with e^{θ_i} the pre-exponential constant A_i . We
 736 set $\theta = (0, 0.3, -0.2)$ and $E = (0.6, 1.1, 0.8)$, and shapes $h_i(x) = 1 + c_i x$ with
 737 $c = (-0.5, 2.0, 0.8)$, giving one decelerating and two accelerating mechanisms.
 738 There is a single input $u \in [0, 3]$, so $m = 1 < d - 1 = 2$, and Theorem 4.1 puts
 739 zero loss out of reach unless this plant is one of the exceptions of Remark 4.5;
 740 Section 6.3 computes the floor and finds that it is not. Noise is unit and the
 741 prior variance is $\text{Var}[R] = 10^3$, so the weight $w = \gamma/(1 + \gamma)$ of Lemma 2.2
 742 is evaluated at the plant’s own $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ at each state rather than
 743 held fixed; γ varies along a trajectory because $\|\mathbf{d}\|$ does. Every trajectory

Table 1: Loss of the fixed indicator at a Fisher score held equal. The duty cycle δ holds $\|\mathbf{d}\|^2$ fixed and leaves $\hat{\mathbf{d}}(\tau)$ unaffected. Losses use $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ with $\text{Var}[R] = 10^3$.

u	δ	$\ \mathbf{d}\ ^2$	rms β (deg)	$\bar{\ell}$ (nat)
0.0	0.235	0.4307565	6.23	0.0215
1.0	0.628	0.4307565	4.53	0.0123
1.5	1.000	0.4307565	6.22	0.0208
2.0	1.564	0.4307565	7.53	0.0379
3.0	3.769	0.4307565	6.57	0.1161

744 starts from $x_i(0) = 5 \times 10^{-3}$ and runs to the condemnation limit $x_f = 0.9$ of
745 (3). Raising x_f to unity leaves every qualitative conclusion below intact and
746 moves the constants: the peak floor of Section 6.3 rises from 4.52° to 5.41° ,
747 and the gain of Section 6.5 falls from 68 to 58. No constant reported below
748 should be read across thresholds.

749 *Verification of the identities.* Before using them we checked the analytical
750 results numerically. The rotation identity (20) holds to relative error below
751 1.1×10^{-7} over the five input-channel pairs $(d, m) = (2, 1), (3, 2), (3, 1),$
752 $(4, 2)$ and $(5, 3)$; the replicator form to 6.9×10^{-8} ; Theorem 3.6 to 5.1×10^{-6} ,
753 limited by finite differencing; and (17) to 1.1×10^{-16} .

754 6.2. A Fisher level set of varying loss

755 Proposition 4.6 is easy to demonstrate on this plant. For a constant
756 input, the duty cycle only reparametrises time, so it moves $\|\mathbf{d}\|$ while leaving
757 $\hat{\mathbf{d}}(\tau)$, and hence β , untouched: the Fisher score and the loss can be varied
758 independently. Table 1 does exactly that, holding the Fisher score equal to
759 seven digits by tuning the duty cycle δ . The loss still spans a factor 9.4.

760 The two objectives also disagree on which input to prefer: the raw score,
761 before δ equalises it, is largest at $u = 0$, which is why δ is smallest there, and
762 the loss is 1.7 times its value at $u = 1$. How far they disagree is a property
763 of the instance; what is structural is only the blindness.

764 6.3. The reachability floor with a single input

765 With $m = 1 < d - 1$ no direction lies on every reachable set at once. Along
766 the closed loop at q^* the floor is largest at the start of life, where $\beta_{\min} =$
767 4.52° ; it falls through mid-life to below the 0.01° resolution of our sampling,
768 where the indicator is momentarily reachable, and rises again towards failure

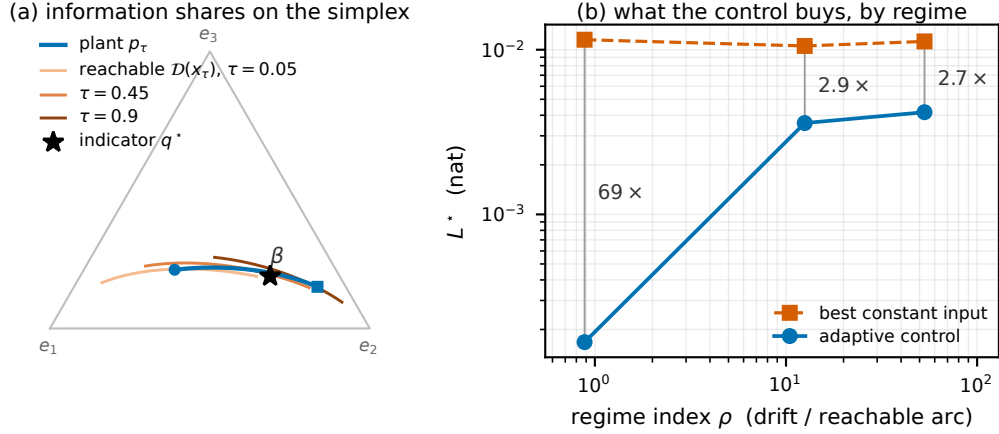


Figure 4: (a) The information-share curve p_τ on the 2-simplex (circle: start, square: failure), the reachable sets $\mathcal{D}(x_\tau)$ at $\tau = 0.05, 0.45$ and 0.9 , and the optimal indicator q^* . No point lies on all three, so β cannot be held at zero (Theorem 4.1). (b) What steering buys, against the regime index ρ .

769 (Fig. 5(a)). Zero average loss would need the indicator reachable at every
 770 instant and not at one, so it is unattainable however the input is steered, in
 771 agreement with Theorem 4.1. Fig. 4(a) shows the geometry: the reachable
 772 set sweeps across q^* as the plant ages, so the mid-life arc passes close to the
 773 indicator while the arcs at $\tau = 0.05$ and $\tau = 0.9$ stand off it.

774 6.4. The counting rule, tested in both directions

775 Theorem 4.1 is an equivalence, and the paragraph above tests only half
 776 of it: with too few inputs the floor stays positive. The other half, that
 777 supplying the inputs the counting rule asks for makes the floor vanish, is the
 778 more useful claim for a practitioner deciding how many operating variables
 779 to instrument, so we test it directly.

780 Give the same three-mechanism plant a second independent input by
 781 writing $\varphi_i(u) = \exp(\theta_i - E_i u_1 - G_i u_2)$ with $G = (0.9, 0.3, 1.2)$, so that $m =$
 782 $2 = d - 1$ and each input acts on the mechanisms in a different pattern.
 783 Then $\log \mathbf{d}_i$ is affine in (u_1, u_2) , and matching a target direction fixes the
 784 $d - 1 = 2$ differences of logarithms: two linear equations in two unknowns,
 785 with a coefficient matrix of determinant 0.27 here, hence a unique solution
 786 at every state.

787 Taking as target the direction the plant occupies at mid-life under a
 788 central operating point, that solution lies inside the admissible box $[0, 3]^2$

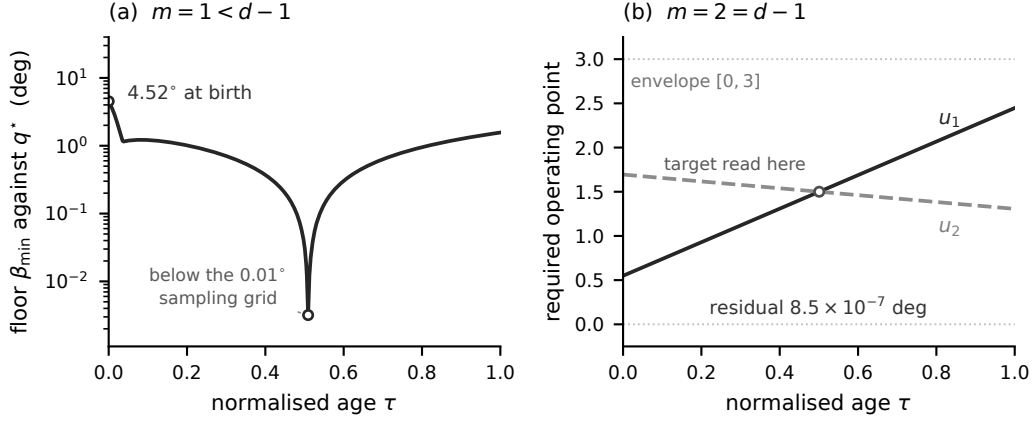


Figure 5: (a) With $m = 1$ the floor against q^* is largest at birth and dips below the sampling grid near mid-life. (b) With $m = 2$ the mid-life direction is held exactly; plotted is the operating point this requires, which never leaves $\mathcal{U}$.

at every one of 600 sampled states covering the whole life, and the residual misalignment there is 8.5×10^{-7} degrees, that is, zero to numerical precision; the schedule it requires is drawn in Fig. 5(b). An independent 90×90 grid search over the box returns a floor of at most 0.156° , which is the resolution of the grid rather than a property of the plant.

The contrast with the single-input case is the content of the theorem: the same mechanisms, the same shapes, the same envelope on each input, and the peak floor moves from 4.52° to zero at every sampled state on the addition of one operating variable. On this plant the counting rule is therefore not a conservative sufficient condition but a sharp one, and within the adopted model it fixes how many independent control degrees of freedom the plant must offer. How those degrees of freedom correspond to physical actuators, and how the mechanisms are individuated in the first place, are modelling questions this paper does not settle. Proposition 4.4 says which plants escape it, and Remark 4.5 says why a designer should not count on being one of them.

6.5. The value of steering

Minimising over q gives $q^* = (0.466, 0.771, 0.435)$, at which the adaptive policy attains 1.68×10^{-4} against 1.15×10^{-2} under the best constant input, a factor of 68. The ratio depends on how finely $\mathcal{U}$ is discretised for the policy, rising to 72 at 161 points; we report it on the 41-point grid used throughout.

810 The factor is an information ratio and is not free in reliability terms:
811 the policy that attains it runs the component differently, and Section 6.6
812 finds the two closed loops separated by a factor of 3.4 in life. Aligning the
813 indicator and prolonging the component are competing objectives, which is
814 what makes the horizon in (24) a controlled quantity rather than a constant.
815 That factor is the value of steering at all, not the value of steering optimally:
816 a myopic policy that attends only to the numerator of (24) sits 2.8% above
817 the optimum at q^* .

818 The fraction in (24) is doing visible work at this operating point. The
819 closed loop at q^* and the closed loop at a balanced indicator, which calls for
820 far less steering, differ by a factor 3.4 in lifetime: what alignment buys in the
821 numerator is paid for in the denominator. This is the tension anticipated in
822 Section 5, and it is why the horizon cannot be fixed in advance.

823 *6.6. The fractional limit*

824 Dinkelbach bisection with the control grid matched to the policy gives,
825 over $N \in \{25, 29, 35\}$, extrapolants agreeing to 1.2% and $\lambda_\infty^* = 1.63 \times 10^{-4}$,
826 below the realisable 1.68×10^{-4} as it must be.

827 We quote this to two digits only, for the following reason: the raw values
828 are twice the extrapolant, so the first-order correction carries half the answer,
829 and both quantities still move by 4% between control grids of 41 and 81
830 points. Three checks guard the number: F must decrease and cross zero, λ^*
831 must not exceed a realisable policy, and the grid sequence must extrapolate
832 consistently. A fixed-horizon formulation failed the second and third and was
833 discarded, as did a first attempt with mismatched control grids. A converse
834 bound uniform over trajectories, which would replace this computation by a
835 theorem, remains open.

836 *6.7. The regime index as a screening test*

837 For a practitioner the first question is not how to steer but whether to
838 bother. One dimensionless ratio answers it. The reachable set $\mathcal{D}(x)$ is an
839 arc; write ρ for the drift of its midpoint over the life divided by its mean
840 length, so ρ measures how far the informative direction wanders in units of
841 the actuation authority available to chase it.

842 Table 2 varies the spread of the activation energies, which by (20) is what
843 sets that authority. As the arc collapses, the attainable loss L^* rises by a
844 factor 25 and the value of control falls from 68 to 2.7: with weak authority
845 there is little left to steer. Fig. 4(b) plots the two against ρ . The construction

Table 2: The regime index. The activation energies E set the actuation authority; c , the noise and $\mathcal{U}$ are held fixed. $L^\star$ is the loss under adaptive control, and the gain its ratio to the best constant input.

E	arc	ρ	$L^\star$	gain
(0.60, 1.10, 0.80)	33.9°	0.88	1.7×10^{-4}	68×
(0.80, 0.82, 0.79)	1.93°	12.5	3.6×10^{-3}	2.9×
(0.80, 0.805, 0.798)	0.45°	53.1	4.2×10^{-3}	2.7×

of this paper is a statement about the low- ρ regime, and ρ can be computed from a nominal degradation model before any indicator is designed.

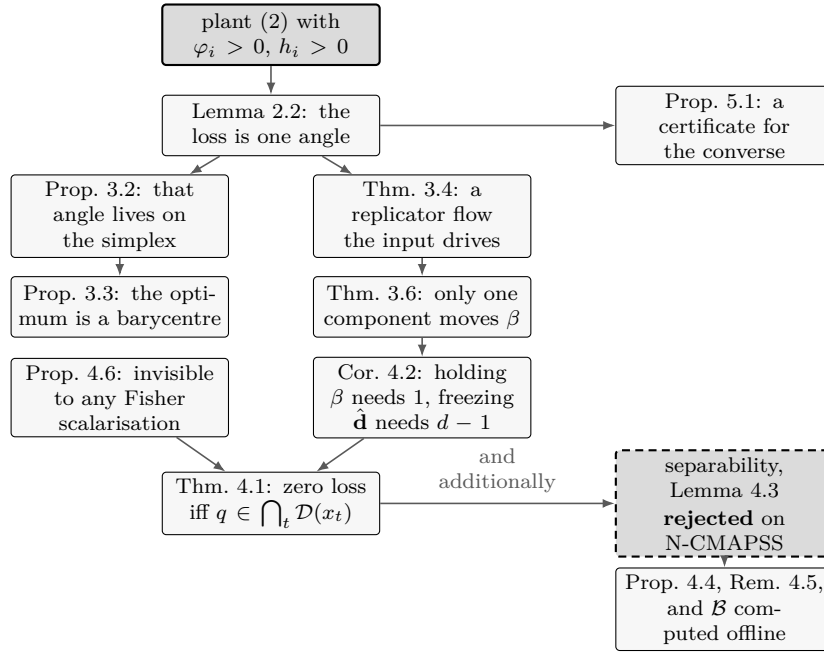
6.8. What the loss buys a downstream estimator

Corollary 2.4 prices the loss in the error of the estimate the model itself prescribes. A practitioner does not use that estimate, so the ordering is worth checking against a procedure that uses none of the construction: sample the indicator over the last 40% of life, match the observed history against the indicator’s own reference curve, and read remaining life off the best-fitting time offset. That estimator sees a scalar signal and nothing else — not $\hat{\mathbf{d}}$, not γ , not the loss.

Over twelve indicators spanning $\bar{\ell}$ from 0.012 to 0.37, its median absolute error — one of the standard prognostic accuracy measures [28] — rises with the loss, at Spearman correlation 0.89, and the best-aligned indicator beats the worst by a factor 1.5. The factor is far smaller than the 31-fold spread in $\bar{\ell}$, and it should be: the loss is a logarithmic information quantity, the error is a time, and an estimator with the whole history recovers much of what any one instant discards. What the experiment settles is the direction, which is what the objection was about.

7. What public data can say about the premises

The construction rests on three modelling choices, and each is empirically testable. They do not carry equal weight, and Fig. 6 separates what depends on which. Two are premises about the plant: that the informative direction rotates along the trajectory, and that the input moves it. The third is the separability of Lemma 4.3, which is what makes the operator’s authority computable offline. This section summarises what six public run-to-failure fleets currently settle: the Severson lithium-ion cells [29], the XJTU-SY [30]



Every result on the spine needs only the positivity of the degradation model. Only the dashed branch needs separability, and it is that hypothesis, not the geometry, that Section 7 reports as rejected on the best public turbofan data.

Figure 6: What rests on what. The spine needs only the positivity of the degradation model. Only the dashed branch adds separability, which Section 7 reports as rejected on the best public turbofan data.

872 and PRONOSTIA [31] bearing fleets, the NASA battery cells [32], and the
 873 C-MAPSS [33] and N-CMAPSS [34] turbofan simulations. The protocols are
 874 not reproduced here; the datasets are named so that each test can be located
 875 and repeated.

876 *Rotation.* On 129 Severson lithium-ion cells [29], taking capacity fade, resis-
 877 tance growth and mean temperature as the three monitoring channels, the
 878 fleet-mean informative direction travels 30.7° of arc over $\tau \in [0.15, 0.9]$ and
 879 ends 21° from where it began (Fig. 2(c)). Mean temperature is a monitoring
 880 channel and not a damage state — it is the operating variable itself in the
 881 Arrhenius form of Section 2 — so this test reads the rotation of a direction
 882 formed from three measured series, which is what the premise asserts, and
 883 does not need (5)’s pairing of channels with mechanisms. The direction
 884 stays in the open positive orthant once the differentiating window reaches

301 cycles, so the positivity that Remark 3.1 shows to be load-bearing is a property of this fleet and not only of the model.

Actuation. This premise needs the operating point to vary while everything else is held fixed, which of the six fleets only the two turbofan simulations permit. On C-MAPSS [33], whose regimes interleave cycle by cycle inside each engine so that an engine serves as its own control, the angle between half-samples of different regimes exceeds the angle between half-samples of the same regime by 7.9° (FD004) and 10.8° (FD002), positive for 436 of 439 engines. Against fake regimes cut from a single real one, where no contrast exists by construction, the same machinery returns $|B - W| \leq 0.5^\circ$. On the four measured fleets the obstacle is the estimator rather than the physics: in the two bearing fleets the operating condition is fixed per unit, so “condition” and “group of units” cannot be separated, and the NASA fleet has four cells. The premise is therefore confirmed on simulation and open on hardware.

Separability. Lemma 4.3 is a statement about a plant class, and data can refute it. If the shift between two operating regimes were the same at every stage of life, regime and stage would act additively on ξ and a two-way layout would show no interaction. On N-CMAPSS [34], which records altitude, Mach number and throttle every second of real flight profiles, the interaction exceeds *both* main effects on the sums of squares, by a factor 1.65 over the operating-point effect on DS02 and 1.84 on DS01, and remains larger under every clustering and staging tried. Separability is therefore rejected on the highest-fidelity public turbofan simulation.

The consequence is the scope of the results above. The geometry of Sections 3 and 4 needs only $\varphi_i > 0$ and $h_i > 0$, and is untouched. What separability buys, and what the rejection withdraws, is Proposition 4.4 and the offline route to $\mathcal{B}$: those describe a model class, and a plant must be shown to belong to it before they are used. The two-way layout above is the test, and it costs one pass over data a reliability programme already holds.

8. Discussion

Four practical readings follow from the results above. The first two hold whatever the plant is, because one is a test and the other an invariance. The last two assume the separable form of (2), which Section 7 reports as rejected on the only data able to test it, so they apply where that form does and not otherwise. The distinction matters more than any of the four.

920 Test the separability assumption before relying on it. The two-way layout
 921 of Section 7 needs only a fleet whose operating point varies within a unit, and
 922 it returns an interaction against a measured noise floor. If the interaction
 923 clears that floor, the stress model is not separable on that fleet, the offline
 924 route to the operator’s authority is unavailable, and the two model-based
 925 readings below do not apply. The test costs one pass over data the programme
 926 already holds.

927 A Fisher criterion cannot be used to design a fixed indicator. This is the
 928 second of the two unconditional readings: Proposition 4.6 is an invariance of
 929 the loss under a common gain, and holds whatever the plant. Design rules
 930 that maximise a scalarisation of the Fisher information, a natural and com-
 931 mon choice, are provably unable to distinguish configurations whose losses
 932 differ by an order of magnitude, because they measure the strength of the
 933 signal and not the direction it points in. If the two objectives coincide in a
 934 given application, that is a property of the application and should be verified,
 935 not assumed.

936 The binding constraint is actuation authority, not indicator sophistica-
 937 tion. The counting rule of Theorem 4.1 prices the operating variables a
 938 lossless fixed indicator would need, and Proposition 4.4 sharpens it: the
 939 count is sufficient and not necessary, but the exceptions ask the plant’s log-
 940 offsets to slide along the input’s own direction, which Remark 4.5 shows to
 941 be a knife edge. Care in choosing the weights substitutes for actuation in
 942 neither case. What one controllable variable does buy is the weaker objective
 943 of Corollary 4.2: an installation that could never freeze the informative direc-
 944 tion can still hold the misalignment constant, though not at zero. Authority
 945 bought this way is not free either. On the test plant the indicator whose loss
 946 is smallest is served by a policy that changes the component’s life by a factor
 947 of 3.4 against a balanced indicator, so information-preserving operation and
 948 long life are competing objectives and the trade between them is a reliability
 949 decision rather than an estimation one.

950 The decision to steer precedes the design of the indicator. The regime
 951 index ρ of Section 6.7 requires only a nominal degradation model and the
 952 operating envelope. When ρ is large, the reachable arc is short relative to
 953 the drift, and Table 2 shows the value of steering collapsing from $68\times$ to
 954 under $3\times$. The index is therefore a screening test to run before committing
 955 engineering effort.

956 *Limitations.* Four limitations qualify the readings above. The separable form
 957 of (2) is not only an idealisation but one the data reject: Section 7 finds the
 958 interaction between operating point and state larger than either main effect
 959 on two N-CMAPSS datasets, so computing the operator’s authority offline
 960 is available only where that form holds. The fundamental limit of Section 5
 961 is computed rather than proved, to two digits. The actuation premise is
 962 confirmed on simulated rather than measured data, for the reasons set out in
 963 Section 7. And the population is taken to slide along one degradation path,
 964 which is what makes remaining life a coordinate on it; under dispersion in
 965 the coefficients of (2) the sensitivity that Remark 2.1 measures stands some
 966 45° from (6), so the construction describes fleets that vary in when they fail
 967 rather than in how they degrade.

968 9. Conclusion

969 A fixed health indicator discards information, and this paper has treated
 970 what it discards as a quantity the operating point controls. Indicator design
 971 then becomes a reachability question. The family of reachable direction sets
 972 bounds what the operating point can achieve, and their drift relative to their
 973 width is a dimensionless measure of whether substantial benefit is plausi-
 974 ble. That ratio is dimensionless and available before the designer fixes any
 975 indicator. Actuation authority, rather than sophistication of the indicator,
 976 governs whether a fixed compression can come close to lossless. Whether
 977 that authority is present is an empirical question rather than a modelling
 978 assumption, and on 439 simulated turbofan records it is. Establishing the
 979 same on measured fleets awaits records in which the operating point varies
 980 enough to show the effect.

981 The separable plant form that makes all of this computable is rejected by
 982 the data, on two datasets of the highest-fidelity public turbofan simulation,
 983 so the geometry of Section 4.1 describes a model class rather than measured
 984 machinery, and the offline route it opens is available only where that class
 985 applies. What replaces it when the stress model is not separable, and whether
 986 a weaker structure still yields a computable reachability criterion, is the
 987 question this leaves.

988 Two further questions remain open, the first of them now in a more
 989 specific form. Proposition 5.1 reduces the search for a converse bound uni-
 990 form over trajectories to the exhibition of a single continuously differentiable
 991 verification function; the value function is the natural candidate and fails

the differential test for a structural reason, so what is required is a smooth under-approximation of it, by sum-of-squares programming or by interval arithmetic with an explicit Lipschitz constant. The other is unchanged: the fractional limit to better than two digits, which requires a scheme of higher order than the semi-Lagrangian one used here.

Data availability and reproducibility

The six fleets named in Section 7 are publicly available. One script regenerates every computed result from a single configuration, and a companion check recomputes each reported number and compares it with the value written in the text, covering the tables and the quantities reported alongside them. Those checks are maintained against the full manuscript from which this paper is drawn, and all of them pass; the numbers reported here are carried over from it unchanged. Three groups of numbers lie outside it: the residuals of the identity checks in Section 6, the barycentre comparison in Section 3.3, and the diagnostics of the two discarded formulations in Section 5. The plots are produced by three further scripts. All of these are available at <https://github.com/HuyHoangLe0201/health-indicator-information-loss>.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

- [1] Y. Lei, N. Li, L. Guo, N. Li, T. Yan, J. Lin, Machinery health prognostics: A systematic review from data acquisition to RUL prediction, *Mech. Syst. Signal Process.* 104 (2018) 799–834. doi:10.1016/j.ymssp.2017.11.016.

- 1022 [2] J. Coble, J. W. Hines, Identifying optimal prognostic parameters from
1023 data: A genetic algorithms approach, in: Annu. Conf. PHM Soc., 2009.
- 1024 [3] H. H. Le, K.-A. Nguyen, Information-theoretic limits of health indicator
1025 construction for remaining useful life prediction, Eng. Sci. Technol. Int.
1026 J. 81 (2026) 102485. doi:10.1016/j.jestch.2026.102485.
- 1027 [4] H. H. Le, K.-A. Nguyen, Generalization bounds and sample complex-
1028 ity for remaining useful life prediction from complete degradation tra-
1029 jectories, Meas. Sci. Technol. 37 (2026) 226203. doi:10.1088/1361-
1030 6501/ae7109.
- 1031 [5] L. Guo, N. Li, F. Jia, Y. Lei, J. Lin, A recurrent neural network based
1032 health indicator for remaining useful life prediction of bearings, Neuro-
1033 computing 240 (2017) 98–109. doi:10.1016/j.neucom.2017.02.045.
- 1034 [6] K. Javed, R. Gouriveau, N. Zerhouni, State of the art and tax-
1035 onomy of prognostics approaches, trends of prognostics applica-
1036 tions and open issues towards maturity at different technology
1037 readiness levels, Mech. Syst. Signal Process. 94 (2017) 214–236.
1038 doi:10.1016/j.ymssp.2017.01.050.
- 1039 [7] Z.-S. Ye, M. Xie, Stochastic modelling and analysis of degradation for
1040 highly reliable products, Appl. Stoch. Models Bus. Ind. 31 (1) (2015)
1041 16–32. doi:10.1002/asmb.2063.
- 1042 [8] J. M. van Noortwijk, A survey of the application of gamma pro-
1043 cesses in maintenance, Reliab. Eng. Syst. Saf. 94 (1) (2009) 2–21.
1044 doi:10.1016/j.ress.2007.03.019.
- 1045 [9] X.-S. Si, W. Wang, C.-H. Hu, D.-H. Zhou, Remaining useful life estima-
1046 tion — a review on the statistical data driven approaches, Eur. J. Oper.
1047 Res. 213 (1) (2011) 1–14. doi:10.1016/j.ejor.2010.11.018.
- 1048 [10] Z. Zhang, X. Si, C. Hu, Y. Lei, Degradation data analysis and remaining
1049 useful life estimation: a review on Wiener-process-based methods, Eur.
1050 J. Oper. Res. 271 (3) (2018) 775–796. doi:10.1016/j.ejor.2018.02.033.
- 1051 [11] J. Lee, F. Wu, W. Zhao, M. Ghaffari, L. Liao, D. Siegel, Prognostics
1052 and health management design for rotary machinery systems — reviews,

- 1053 methodology and applications, *Mech. Syst. Signal Process.* 42 (1–2)
1054 (2014) 314–334. doi:10.1016/j.ymssp.2013.06.004.
- 1055 [12] F. Karimi Pour, D. Theilliol, V. Puig, G. Cembrano, Health-
1056 aware control design based on remaining useful life estimation
1057 for autonomous racing vehicle, *ISA Trans.* 113 (2021) 196–209.
1058 doi:10.1016/j.isatra.2020.03.032.
- 1059 [13] M. S. Félix, J. J. Martinez, C. Bérenguer, Remaining useful life (RUL)
1060 control of controlled systems under degradation, *Int. J. Robust Nonlin-*
1061 *ear Control* 35 (18) (2025) 8095–8116. doi:10.1002/rnc.70105.
- 1062 [14] A. A. Feldbaum, Dual control theory I, *Autom. Remote Control* 21 (9)
1063 (1960) 874–880.
- 1064 [15] C. Grebe, E. Wise, J. Kelly, Observability-aware trajectory op-
1065 timization: Theory, viability, and state of the art, in: *IEEE*
1066 *Int. Conf. Multisensor Fusion Integr. (MFI)*, 2021, pp. 1–8.
1067 doi:10.1109/mfi52462.2021.9591177.
- 1068 [16] G. Hooker, K. K. Lin, B. Rogers, Control theory and experimental design
1069 in diffusion processes, *SIAM/ASA J. Uncertain. Quantif.* 3 (1) (2015)
1070 234–264. doi:10.1137/140962280.
- 1071 [17] H. Lim, B.-J. Yum, Optimal design of accelerated degradation tests
1072 based on Wiener process models, *J. Appl. Stat.* 38 (2) (2011) 309–325.
1073 doi:10.1080/02664760903406488.
- 1074 [18] S.-i. Amari, *Information Geometry and Its Applications*, Springer, 2016.
1075 doi:10.1007/978-4-431-55978-8.
- 1076 [19] A. Bhattacharyya, On a measure of divergence between two statistical
1077 populations defined by their probability distributions, *Bull. Calcutta*
1078 *Math. Soc.* 35 (1943) 99–109.
- 1079 [20] J. Hofbauer, K. Sigmund, *Evolutionary Games and Population Dynam-*
1080 *ics*, Cambridge Univ. Press, 1998. doi:10.1017/cbo9781139173179.
- 1081 [21] W. H. Sandholm, *Population Games and Evolutionary Dynamics*, MIT
1082 Press, 2010.

- 1083 [22] W. Q. Meeker, L. A. Escobar, Statistical Methods for Reliability Data,
1084 Wiley, 1998.
- 1085 [23] T. M. Cover, J. A. Thomas, Elements of Information Theory, 2nd Edi-
1086 tion, Wiley, 2006. doi:10.1002/047174882x.
- 1087 [24] E. Oja, A simplified neuron model as a principal component analyzer,
1088 J. Math. Biol. 15 (1982) 267–273. doi:10.1007/bf00275687.
- 1089 [25] M. Bardi, I. Capuzzo-Dolcetta, Optimal Control and Viscosity So-
1090 lutions of Hamilton–Jacobi–Bellman Equations, Birkhäuser, 1997.
1091 doi:10.1007/978-0-8176-4755-1.
- 1092 [26] W. Dinkelbach, On nonlinear fractional programming, Manage. Sci.
1093 13 (7) (1967) 492–498. doi:10.1287/mnsc.13.7.492.
- 1094 [27] M. Falcone, R. Ferretti, Semi-Lagrangian Approximation Schemes
1095 for Linear and Hamilton–Jacobi Equations, SIAM, 2014.
1096 doi:10.1137/1.9781611973051.
- 1097 [28] A. Saxena, J. Celaya, E. Balaban, K. Goebel, B. Saha, S. Saha,
1098 M. Schwabacher, Metrics for evaluating performance of prognostic tech-
1099 niques, in: International Conference on Prognostics and Health Man-
1100 agement, 2008, pp. 1–17. doi:10.1109/phm.2008.4711436.
- 1101 [29] K. A. Severson, P. M. Attia, N. Jin, N. Perkins, B. Jiang, Z. Yang, M. H.
1102 Chen, M. Aykol, P. K. Herring, D. Fraggdakis, M. Z. Bazant, S. J.
1103 Harris, W. C. Chueh, R. D. Braatz, Data-driven prediction of battery
1104 cycle life before capacity degradation, Nat. Energy 4 (2019) 383–391.
1105 doi:10.1038/s41560-019-0356-8.
- 1106 [30] B. Wang, Y. Lei, N. Li, N. Li, A hybrid prognostics approach for es-
1107 timating remaining useful life of rolling element bearings, IEEE Trans.
1108 Rel. 69 (1) (2020) 401–412. doi:10.1109/tr.2018.2882682.
- 1109 [31] P. Nectoux, R. Gouriveau, K. Medjaher, E. Ramasso, B. Chebel-Morello,
1110 N. Zerhouni, C. Varnier, PRONOSTIA: An experimental platform for
1111 bearings accelerated degradation tests, in: IEEE Int. Conf. Prognostics
1112 Health Manage. (PHM), Denver, CO, USA, 2012, pp. 1–8.

- 1113 [32] B. Saha, K. Goebel, Battery data set, NASA Ames Prognostics Data
1114 Repository (2007).
- 1115 [33] A. Saxena, K. Goebel, D. Simon, N. Eklund, Damage propagation
1116 modeling for aircraft engine run-to-failure simulation, in: Int. Conf.
1117 Prognostics Health Manage. (PHM), Denver, CO, USA, 2008, pp. 1–
1118 9. doi:10.1109/phm.2008.4711414.
- 1119 [34] M. Arias Chao, C. Kulkarni, K. Goebel, O. Fink, Aircraft engine run-
1120 to-failure dataset under real flight conditions for prognostics and diag-
1121 nostics, Data 6 (1) (2021) 5. doi:10.3390/data6010005.